

# Lecture 06: Evaluation of evaluations

**PPHA 34600**  
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# From last time: RCTs are the bee's knees

Randomized controlled trials are super powerful:

- Random assignment allows us to solve our selection problem
- We can implement them with tweaks to handle challenges:
  - *Noncompliance*: Dividing  $\tau^{ITT}$  by share of compliers  $\rightarrow \tau^{LATE}$
  - *Spillovers*: Proper design to avoid or measure
- More opportunities than you might imagine for implementation

# Moving out of RCT land

We will spend the rest of the course on other research designs:

- Randomized controlled trials (RCTs)
- Trying to control for observable things
- Panel data
- Instrumental variables
- Regression discontinuity
- Big Data and machine learning

# Why leave RCT land?

RCTs are the gold standard for a reason, but:

- They can be expensive
- Some programs require evaluation at scale
- RCTs can't always be implemented
- **There's a lot to learn from non-RCTs**

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- “What experiment would I run to answer this question?”
- Useful to nail down your question of interest
- Valuable to think through problems with your non-RCT

# The ideal experiment

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**ideal experiment**

- “What experiment would I run to answer this question?”
- Can be totally feasible (RED for energy efficiency upgrades)...
- ...or totally infeasible (randomly warm one Earth while keeping the other cold)

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- Use quasi-experimental methods to estimate  $\hat{\tau}$
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This is often referred to as a “LaLonde exercise” after LaLonde (1986)

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**Context is really important for this!**



# Leveraging an RCT we know and love

*Blast from the past: We'll use the SMUD pricing RCT*

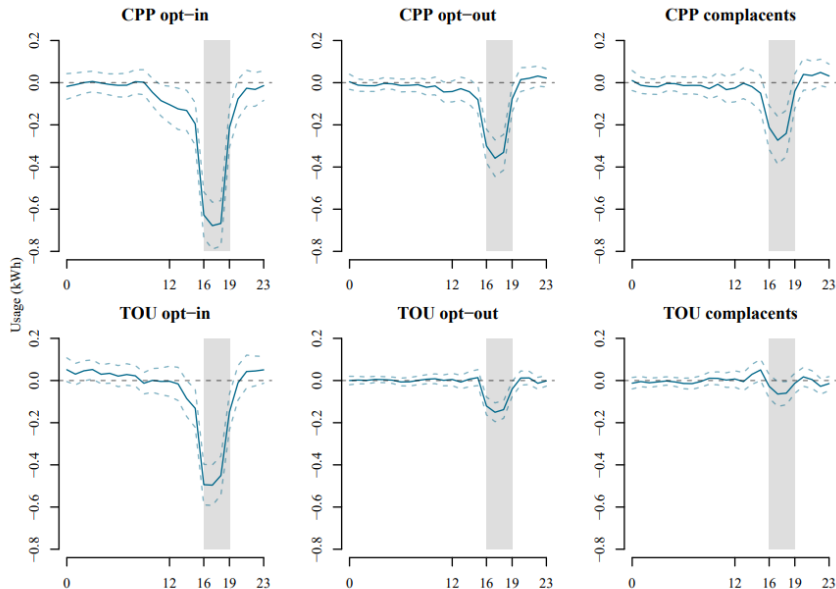
## Policy issue:

- The cost of providing electricity is time-varying
- Prices typically aren't
- This causes large welfare losses

## Program:

- SMUD (randomly) implemented time-varying pricing
- Experimental run: 2011-2013
- Two flavors: “time-of-use” (TOU) and “critical peak pricing” (CPP)
- Both opt-in and opt-out versions

# Fowlie & Wolfram et al results recap



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### Potential for selection into treatment on $\tau_i$

- People who choose to get treated may have different price sensitivity
- We know some of this is happening! (two LATEs)

# What would the naive estimator do?

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**Why is this problematic for electricity pricing?**

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A research design:

- Tries to solve the selection problem without randomization
- Invokes stronger assumptions than the RCT
- Allows us to make progress without randomization
- Best-case scenario: mimics an RCT

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## 3 Regression discontinuity

- Essentially compares just-treated units to just-untreated units
- Leverages cutoffs in policy

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  - Why do this?  
→ Control for common shocks to everyone
- 3 Subtract difference (1) from difference (2):

$$y_{it} = \alpha + \tau D_{it} + \underbrace{\gamma_i}_{i \text{ to itself}} + \underbrace{\delta_t}_{i \text{ to } j \text{ over time}} + \varepsilon_{it}$$

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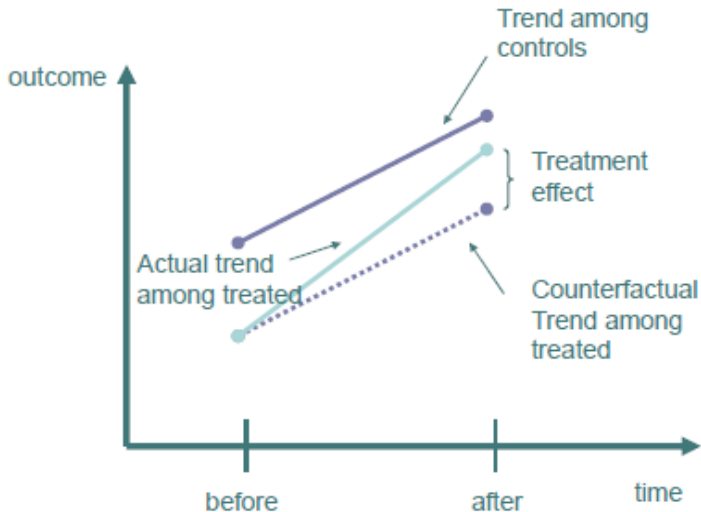
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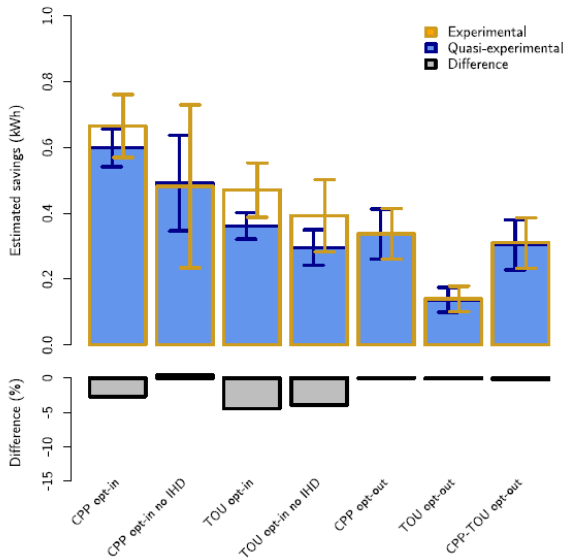
Note: Spurlock et al drop encouraged-but-untreated units

→ Was this necessary?

# Difference in difference intuition



# Comparing experimental and diff-in-diff results



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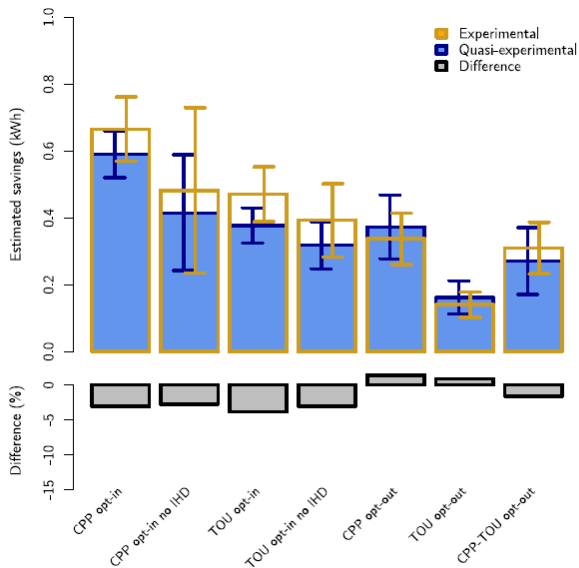
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For this to work, we require:

- Our selection control soaks up everything that matters!

Note: Same approach as DD, but now controlling for more

# Comparing experimental and propensity score results



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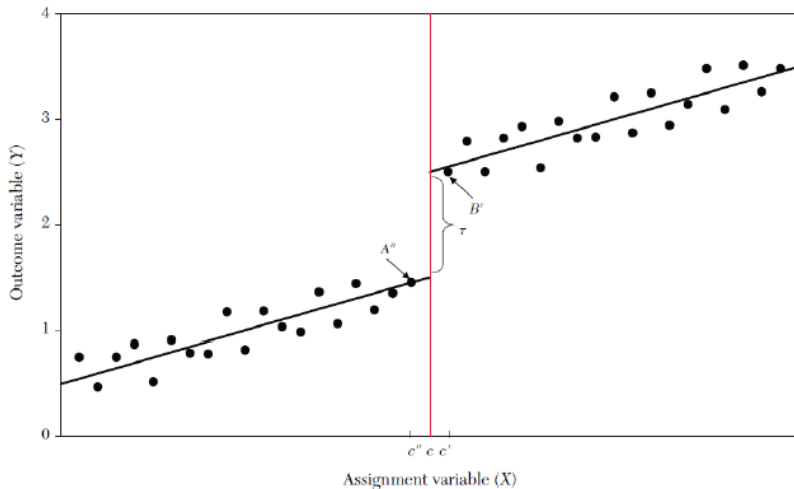
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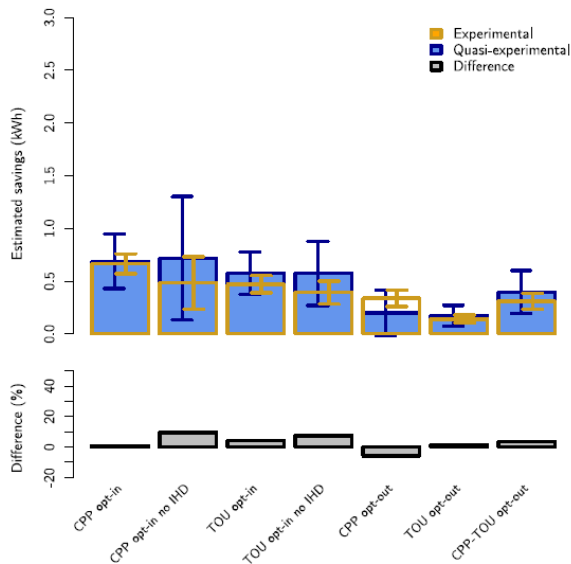
Note: Spurlock et al construct fake cutoffs

- They stitch together control group units with treatment group units
- The stitching point is their artificial cutoff

# Regression discontinuity intuition

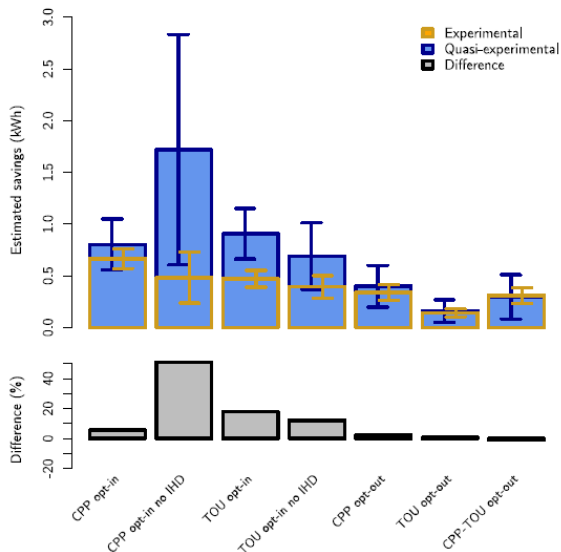


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  - OR estimating a different LATE
- Opt-out treatments are less biased than opt-in treatments
  - Intuition: We do better with a less-selected treatment

## Exercise caution with non-experimental results

*“Even though I was unable to evaluate all non-experimental methods, this evidence suggests that policymakers should be aware that the available non-experimental evaluations...may contain large and unknown biases resulting from specification errors.” – LaLonde (1986)*

## TL;DR:

- ① RCTs are (still) great!
- ② Quasi-experimental methods can get things wrong
- ③ We don't usually have a good experimental benchmark ()

# For next class

## Topics:

- Selection on observables

Reading: Davis, Fuchs, and Gertler (2014). You can skip:

- IV: Mechanisms
- V: Cost effectiveness